

```
1  /*Importing the Dataset*/
2  Filename Reffile"/home/u64454765/MM711 Programming with SAS/Data/Characteristics.xlsx";
3  proc import datafile=Reffile
4      DBMS=XLSX
5      Out=Char;
6      Getnames=Yes;
7  Run;
8  Filename Reffile"/home/u64454765/MM711 Programming with SAS/Data/Houseprice.xlsx";
9  proc import datafile=Reffile
10     DBMS=XLSX
11     Out=HousePrice;
12     Getnames=Yes;
13  Run;
14
15  /*Remove Duplicates*/
16  proc sort data=HousePrice out=prices_unique nodupkey;
17      by Reference;
18  run;
19  proc sort data=char out=chars_sorted;
20      by Reference;
21  run;
22
23  /*Checking Missing Values*/
24  proc means data=char nmiss;
25  title "Missing Values in Housing Characteristics";
26  Run;
27  proc means data=HousePrice nmiss;
28  title "Missing Values in House Price";
29  run;
30  proc freq data=Char;
31  title "Missing Garage Types";
32  table Garage_type;
33  Run;
34
35  /*Merging & Cleaning the Datasets*/
36  data house_clean;
37      merge chars_sorted(in=a) prices_unique(in=b);
38      by Reference;
```

```
40     if Garage_Type = "NA" then Garage_Type = "";
41     if SoldPrice = . or YearSold = . then delete;
42     if LivingArea_sm <= 0 or LivingArea_sm > 500 then delete;
43     HouseAge = YearSold - ConstructionYear;
44 run;
45
46 /*Removing Extreme Values for Final Data*/
47 data Final;
48 set house_clean;
49 HouseAge = YearSold - ConstructionYear;
50 where livingArea_sm lt 1000;
51 Run;
52
53 /* Test for normality */
54 proc univariate data=WORK.FINAL normal mu0=0;
55     ods select TestsForNormality;
56     var SoldPrice;
57 run;
58
59 /* t test */
60 proc ttest data=WORK.FINAL sides=2 h0=0 plots(showh0);
61     var SoldPrice;
62 run;
63
64 /*Correlation Analysis*/
65 ods noproctitle;
66 ods graphics / imagemap=on;
67 proc corr data=WORK.FINAL pearson nosimple noprob plots=none;
68     var LivingArea_sm Bedroom Bathroom HouseAge;
69     with SoldPrice;
70 run;
71
72 /*Visualization*/
73 proc sgplot data=Final;
74     title "Impact of Living Area on Sold Price";
75     scatter x=LivingArea_sm y=SoldPrice;
76     reg x=LivingArea_sm y=SoldPrice / degree=1; /* Adds a trend line */
77 run;
```

```
80 proc sgplot data=Final;
81     title "Sold Price Distribution by Number of Bedrooms";
82     vbox SoldPrice / category=Bedroom;
83 run;
84
85 proc sgplot data=Final;
86     title "Distribution of Property Sale Prices";
87     histogram SoldPrice;
88     density SoldPrice; /* Adds a distribution curve */
89 run;
90
91 /*Regression*/
92 proc reg data=Final;
93     title "Multiple Linear Regression: Predicting Sold Price";
94     model SoldPrice = LivingArea sm Garage sm Bedroom Bathroom HouseAge;
```